

PRODUCT REQUIREMENTS DOCUMENT

AI Mentor

Adaptive AI Learning Platform, Roadmap Engine, and Context-Aware Mentoring System

Document Value	Decision
Version	1.0 - Consolidated Product and Infrastructure Master Plan
Status	Implementation baseline for MVP; architecture and AI-provider details remain ADR-controlled
Initial market	General self-directed learners; Arabic and English first-class
Primary customer	Students, employees, professionals, freelancers, career switchers, and upskilling/reskilling learners
Commercial model	MVP is fully free; paid plans are explicitly deferred
Product owner	Founder / Product Owner
Prepared	20 August 2026

Document purpose: This document is the single product and technical baseline for discovery, design, estimation, implementation, AI integration, quality assurance, MVP release, and controlled future expansion. It preserves the broader product vision while explicitly separating MVP scope from later capabilities.

Document Control

Item	Decision
Product name	AI Mentor; working product name may be rebranded without changing domain terminology.
Source baseline	Consolidates the supplied AI Mentor prototype, confirmed Product Owner decisions in requirements discovery, and this engineering analysis.
Requirements language	MUST = mandatory, SHOULD = recommended, MAY = optional or configurable.
AI authority	AI may generate, explain, assess, recommend, and propose changes. Material roadmap changes require explicit user approval.
Source of truth	Persisted application data is authoritative. AI output is never treated as durable state until validated and stored.
Change control	Material changes to roadmap lifecycle, AI authority, progress rules, data model, security, privacy, or resource trust require a recorded Product or Architecture Decision Record.

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1. Executive Summary

AI Mentor is a general-purpose adaptive learning platform that converts a learner-defined goal into a personalized, staged roadmap and supports execution through structured tasks, trusted external resources, progress tracking, and a context-aware AI mentor. The product is intentionally domain-agnostic: a learner may pursue programming, design, marketing, languages, business, data, public speaking, or another self-directed learning objective without the system being hard-coded to one curriculum.

The MVP focuses on a single active roadmap per user. The AI generates the roadmap from the learner profile, stores the generated structure as durable application data, and then assists the learner using a concise context consisting primarily of goal, current stage, current task, progress, and relevant preferences. The AI may suggest roadmap changes, but it does not silently mutate the learner's plan; changes flow through explicit proposals that the learner accepts or rejects.

Product strategy: establish trustworthy roadmap generation, resource quality, predictable state transitions, and useful context-aware mentoring before adding template libraries, multiple concurrent roadmaps, graded assessments, monetization, notifications, or advanced social/community features.

1.1 Core Value Proposition

- Turn a vague learning goal into a clear sequence of stages and actionable tasks.
- Personalize pace and content using learner level, preferred learning methods, available time, target outcome, and resource language.
- Reduce search overload by attaching verified, relevant, recent, and trustworthy external resources directly to tasks.
- Provide one AI mentor that stays aware of where the learner currently is instead of behaving like a generic chat assistant.
- Keep the learner in control: AI proposes meaningful roadmap changes and the learner approves or rejects them.
- Make progress visible through task completion, stage completion, streaks, milestones, achievements, and timeline history.

1.2 Product Components

Component	Primary users	Purpose
Learning Web Application	Registered learners	Onboarding, roadmap, tasks, resources, mentor chat, progress, settings, and lifecycle actions.
AI Roadmap Engine	Learners / backend services	Generate a structured roadmap from learning goal and profile; regenerate before learning begins.
AI Mentor Engine	Learners	Answer context-aware questions, explain concepts, assess progress signals, recommend resources, and propose plan changes.
Resource Discovery & Validation	AI orchestration / backend workers	Find, rank, verify, normalize, and periodically re-check external learning resources.
Progress & Gamification	Learners	Track task/stage completion, streaks, milestones, achievements, and progress history.
Platform Administration	Administrators / support operators	User support, AI usage visibility, content/resource governance, safety controls, and operational oversight; built after core learner flows.

2. Product Vision and Business Strategy

2.1 Problem Statement

- Learners often know what they want to achieve but do not know the correct sequence of topics, prerequisites, or practice activities.
- Generic search and generic chat answers create information overload instead of a stable learning plan.
- Online learning resources vary widely in freshness, credibility, popularity, language, difficulty, and continued availability.
- Fixed courses do not adapt well when a learner is ahead, behind, changes preferences, or becomes stuck on one task.
- Learners lose momentum when progress is invisible or when a single unsuitable task blocks the entire plan.
- General-purpose AI assistants often lack durable learning context and therefore repeat questions, forget progress, or recommend work that conflicts with the current roadmap.

2.2 Target Users

Segment	Priority	Primary need
Self-directed beginner	Launch priority	A clear starting point, manageable tasks, explanations, and trusted resources.
Student	Launch priority	Structured skill development outside or alongside formal study.
Employee / professional	Launch priority	Upskilling and reskilling with realistic time

Segment	Priority	Primary need
		constraints.
Career switcher	Launch priority	Outcome-oriented roadmap toward a new role or demonstrable skill.
Freelancer / independent learner	Supported	Targeted skill acquisition with project-oriented work.
Organizations / teams	Future	Shared learning programs, cohorts, analytics, assignments, and administration.

2.3 Commercial Model

- The MVP is fully free.
- No payment gateway, subscription enforcement, usage quota, credit system, or premium entitlement is required for MVP.
- The architecture SHOULD record AI usage and cost metrics even while the product is free, so later commercial models can be evaluated with real data.
- Future monetization MAY include AI usage tiers, multiple roadmaps, premium model access, advanced analytics, curated templates, organization plans, certificates, or mentor-review capabilities.

2.4 Differentiation Strategy

- Roadmap-first rather than chat-first product design.
- Context-aware mentoring tied to persisted roadmap state.
- User-approved adaptation rather than silent AI plan mutation.
- External-resource quality controls instead of unverified link generation.
- General-purpose learning domain with structured task taxonomy and outcome-oriented progression.
- Arabic RTL and English LTR as first-class product experiences.

3. Scope and Release Strategy

3.1 Full Product Scope

- Account registration, login, password reset, session management, profile, account deletion, and bilingual interface.
- Learning onboarding: goal, self-assessed level, learning preferences, available time, desired outcome, and preferred resource language.
- AI-generated roadmap with stages, tasks, dependencies/order, estimated effort, milestones, and achievements.
- Task types: Read, Watch, Quiz, Project, Assignment, and Coding Challenge; future types may be added without schema redesign.
- Task completion, skip workflow, replacement suggestions, stage completion, roadmap progress, streaks, timeline, and achievements.
- External resource discovery, ranking, verification, deduplication, freshness checking, and task association.
- One context-aware mentor chat per user with dynamic context based on current product screen and roadmap state.
- AI-generated roadmap adaptation proposals requiring explicit user accept/reject.
- Roadmap regeneration before learning begins; roadmap reset; goal-change rebuild behavior; roadmap archive/history-ready model.
- Administrator capabilities for support, AI operations, abuse handling, resource governance, and observability after the core learner MVP is stable.

3.2 Release Phases

Phase	Target	Included outcome
Phase 0 - Product & AI Foundation	2-3 weeks	Final UX flows, ADRs, prompt contracts, provider abstraction, data model, resource trust policy, evaluation dataset, CI/CD baseline.
Phase 1 - Core MVP Alpha	6-8 weeks	Authentication, onboarding, one-roadmap lifecycle, AI generation, stages/tasks, task

Phase	Target	Included outcome
		completion, progress basics, bilingual UI.
Phase 2 - Mentor & Resource Beta	4-6 weeks	Context-aware chat, resource discovery/validation, skip/replacement flow, adaptation proposals, milestones, achievements, streaks, hardening.
Phase 3 - Controlled Public MVP	3-4 weeks	Operational dashboards, support tools, AI cost/quality monitoring, security/privacy review, load testing, closed beta, defect correction.
Phase 4 - Growth	Later	Multiple roadmaps, templates, graded assessments, notifications, monetization, organization learning, deeper analytics, mobile app if validated.

Indicative MVP duration: approximately 13-18 weeks for a production-capable first release with a focused 2-3 developer team plus Product Owner involvement, assuming stable scope and available AI/search provider access. Duration changes materially if provider integrations, web-resource verification, or UX scope expand.

3.3 Explicit Non-Goals for MVP

- Multiple concurrent active roadmaps per user.
- Curated human-authored roadmap/template marketplace.
- Mandatory quizzes or automated proof-of-work before ordinary task completion.
- AI grading of uploaded assignments as a completion gate.
- Calendar scheduling, automatic rescheduling after missed days, or time-block planning.
- Push, email, SMS, or WhatsApp learning reminders.
- Paid subscriptions, payment gateways, credit packs, or premium feature enforcement.
- Social feed, cohorts, leaderboards, peer messaging, or community moderation.
- Native mobile applications.
- Offline learning-state mutation and synchronization.
- Enterprise SSO, team administration, or organization analytics.

4. Users, Roles, and Product Boundaries

4.1 User Model

Role / actor	Core responsibilities
Learner	Own account, define learning profile, generate/regenerate/reset roadmap, complete/skip tasks, use mentor, accept/reject proposals, manage preferences and account.
AI Roadmap Engine	Generate structured roadmap and metadata from validated learner inputs.
AI Mentor	Answer contextual questions, explain concepts, recommend resources, evaluate signals, and create change proposals.
Resource Discovery Service	Discover candidate external resources and return evidence used for ranking and validation.
Resource Validation Service	Check URL accessibility, metadata, source trust, duplication, language, recency, and task relevance.
Platform Administrator	Support users, manage provider/configuration controls, inspect quality/usage, disable unsafe integrations, and manage operational incidents.

4.2 MVP Ownership Model

- Each user MUST own at most one active roadmap in MVP.

- The data model **MUST NOT** assume one-roadmap-per-user permanently; roadmap records remain one-to-many capable so multiple roadmaps can be enabled later without structural migration.
- Chat is logically one conversation stream for the learner in MVP; message context is dynamically enriched from the screen and current roadmap state.
- Roadmap-generated content belongs to the learner account and remains stored until roadmap reset/deletion policy is applied.

4.3 Authorization Principles

- Learners can access only their own roadmap, tasks, resources, chat history, progress, and proposals.
- Administrative support access to learner content **MUST** be permission controlled and auditable.
- AI provider credentials, prompt templates, model configuration, moderation rules, and operational logs are platform-only assets.
- UI visibility does not replace backend ownership checks.

5. Core Business Rules

ID	Rule	Definition
BR-001	Single active roadmap	A learner may have at most one active roadmap in MVP. Historical/archive-ready records MAY exist internally for future expansion.
BR-002	Roadmap source of truth	Validated persisted roadmap data is authoritative. An AI response does not change application state until the backend validates and commits the change.
BR-003	AI-first generation	MVP roadmaps are generated by AI from the learner profile rather than selected from a template catalog.
BR-004	Learner-controlled adaptation	The AI may propose roadmap changes but MUST NOT silently apply material additions, removals, replacements, reordering, level changes, or restructuring.
BR-005	Self-assessed level	MVP level is selected by the learner; no placement examination is required.
BR-006	Task taxonomy	Every task has a supported task type. Initial types are Read, Watch, Quiz, Project, Assignment, and Coding Challenge.
BR-007	Completion authority	Ordinary task completion is learner-declared by Mark as Complete. No proof, exam, upload, or AI grading is mandatory in MVP.
BR-008	Equal task weighting	Each task contributes one equal unit to roadmap and stage completion calculations in MVP.
BR-009	Stage completion	A stage becomes Complete only when all non-replaced, non-removed required tasks in that stage are complete.
BR-010	Skip workflow	A learner may request to skip a task. The mentor SHOULD ask for a reason and recommend Keep, Replace, or Skip before final action.
BR-011	Streak	A calendar day counts toward the learning streak when the learner completes at least one task that day.
BR-012	No missed-day rescheduling	Missing one or more days does not automatically change task order, roadmap dates, or roadmap structure in MVP.
BR-013	Time preference	Available time is a pace input used to estimate task size and roadmap duration; it is

ID	Rule	Definition
		not a calendar schedule.
BR-014	Resource trust	Resources shown as recommended MUST pass defined validity and quality checks; the AI must not expose arbitrary unverified links as trusted recommendations.
BR-015	Resource freshness	The platform SHOULD prefer recent and maintained resources when subject matter changes rapidly, while allowing authoritative older foundational material when it remains current.
BR-016	Resource language	Learners choose Arabic, English, or Both for preferred resource language independently from UI language.
BR-017	Context minimization	Mentor requests use concise relevant context rather than indiscriminately sending the full application history.
BR-018	Single chat stream	MVP maintains one mentor chat stream per learner; the active screen determines additional context injected for a request.
BR-019	Goal change	Changing the learning goal after roadmap creation invalidates the current roadmap and triggers rebuild behavior rather than partial semantic merging.
BR-020	Regeneration window	A learner may regenerate a roadmap before learning begins. After learning begins, material changes SHOULD occur through explicit adaptation proposals or reset/rebuild.
BR-021	Reset roadmap	Reset removes the active learning plan from active use and allows a new roadmap to be created; retention behavior is controlled by deletion policy.
BR-022	Achievements	Milestones and achievements are generated dynamically from roadmap content and learner progress, not limited to one fixed global list.
BR-023	MVP pricing	All MVP learner functionality is free and no premium entitlement check blocks core learning flows.
BR-024	Bilingual product	Arabic RTL and English LTR are first-class. UI language and learning-resource language are separate preferences.
BR-025	Account deletion	A learner may request account deletion; personal data and owned learning data are deleted or anonymized according to the platform retention policy.

6. End-to-End Workflows

6.1 Registration and First Roadmap Setup

1. Learner registers with name, email, and password and enters the authenticated application.
2. System collects learning goal, self-assessed level, learning preferences, available time, desired outcome, and preferred resource language.
3. Backend validates required inputs and stores the learner profile.
4. AI Roadmap Engine receives a structured generation contract and returns roadmap metadata, ordered stages, tasks, milestones, and initial resource intents.
5. Backend validates schema, limits, task types, ordering, and required fields before persistence.

6. System stores the roadmap and redirects the learner to the dashboard.

6.2 Roadmap Generation and Regeneration

7. Learner submits onboarding inputs or requests Regenerate before starting the roadmap.
8. System records a generation request with prompt/model/provider metadata and a correlation ID.
9. AI returns structured output; invalid output is repaired/retried within policy rather than partially persisted.
10. Resource intents are handed to the resource pipeline for discovery and verification.
11. Only after validation does the roadmap become Ready.
12. If regenerated before learning starts, the prior draft/generated roadmap is superseded and retained only according to history policy.

6.3 Daily Learning Task

13. Learner opens Dashboard and sees the current stage, next actionable task, progress, and mentor suggestion.
14. Learner opens the task detail page and reviews objective, estimated time, difficulty, instructions, and attached verified resources.
15. Learner performs the work externally or inside the product depending on task type.
16. Learner marks the task complete.
17. System writes completion history, updates progress projection, evaluates stage completion, records streak activity, and unlocks the next task/stage as applicable.
18. Dashboard and mentor context immediately reflect the new state.

6.4 Skip or Replace Task

19. Learner selects Skip on the current or accessible task.
20. System asks for an optional/required reason according to UX policy and sends task context to the mentor.
21. Mentor recommends Keep, Replace, or Skip and explains the learning impact.
22. Learner chooses the final action.
23. If Replace is chosen, AI proposes a replacement task and resources; backend validates and persists the replacement while preserving change history.
24. If Skip is confirmed, task state and progress rules are updated according to skip policy without pretending the task was completed.

6.5 Context-Aware Mentor Chat

25. Learner opens Mentor directly or enters it from Dashboard, Roadmap, Task, Resource, or Progress.
26. Client sends the user message plus context origin.
27. Backend Context Builder loads compact learner/roadmap context and screen-specific data.
28. AI Mentor receives system policy, learner message, structured context, and allowed tool capabilities.
29. Mentor returns an answer and, when appropriate, typed actions or a roadmap change proposal.
30. Chat message and AI metadata are persisted; state-changing proposals remain Pending until learner approval.

6.6 Roadmap Adaptation Proposal

31. AI detects a reason to change future learning work or the learner explicitly asks for a modification.
32. System creates a proposal containing change type, reason, affected entities, before/after values, and user-facing explanation.
33. Learner reviews the proposal and chooses Accept or Reject.
34. Accept applies the validated mutation transactionally and records applied_at plus resulting version.
35. Reject closes the proposal without modifying the roadmap.
36. All proposal decisions remain traceable for product analytics and support.

6.7 Resource Discovery and Verification

37. Roadmap/task generation produces a resource need rather than blindly trusting a generated URL.
38. Discovery adapter searches supported providers/web sources using task topic, level, language, format, and freshness criteria.
39. Candidates are normalized, deduplicated, classified, and scored.
40. Validation checks URL reachability, source identity, language, relevance, publication/update signals where available, and trust criteria.
41. Highest-scoring valid resources are attached to the task with evidence metadata and last_verified_at.

42. Background revalidation may mark stale/broken resources and trigger replacement discovery.

6.8 Goal Change and Roadmap Rebuild

43. Learner edits the primary learning goal after a roadmap exists.
44. System warns that the current roadmap will be replaced rather than partially merged.
45. After confirmation, current roadmap is archived/reset according to retention policy.
46. Learner profile is updated and a new AI roadmap generation starts.
47. New roadmap progress begins independently; old progress does not artificially inflate the new roadmap.

6.9 Account Deletion

48. Learner requests account deletion and re-authenticates when required.
49. System explains irreversible impact and applicable retention window.
50. Active sessions are revoked.
51. Personal profile, roadmap, chat, progress, and proposal data are deleted or anonymized according to policy; operational/security evidence is retained only where justified.
52. External AI providers are not treated as the system of record and provider-side retention configuration is reviewed separately.

7. Functional Requirements

Requirements are grouped by domain. Phase values indicate earliest intended activation. P0 = required for release correctness; P1 = important for MVP quality; P2 = deferrable or future-ready.

7.1 Identity, Access, and Account

IAM-001 - User registration

Phase: 1 | Priority: P0

The system **MUST** allow a learner to create an account using supported identity fields.

- Email uniqueness is validated case-insensitively after normalization.
- Password policy is enforced server-side.
- Registration success establishes an authenticated session or verified-login path according to auth policy.

IAM-002 - User login

Phase: 1 | Priority: P0

The system **MUST** authenticate learners using secure, revocable sessions.

- Failed attempts are rate limited.
- Sessions support expiry, refresh/rotation where applicable, logout, and revocation.
- Authentication events are security logged without storing secrets.

IAM-003 - Password reset

Phase: 1 | Priority: P0

The system **MUST** support secure password reset through a time-limited, single-purpose recovery flow.

- Reset tokens are one-time and expire.
- Successful reset invalidates prior sessions where policy requires.

IAM-004 - Current user profile

Phase: 1 | Priority: P0

Authenticated learners **MUST** view and update supported profile fields and preferences.

- Ownership checks are enforced server-side.
- Profile updates do not silently rewrite historical roadmap snapshots.

IAM-005 - Account deletion

Phase: 2 | Priority: P0

A learner **MUST** be able to request permanent account deletion.

- Re-authentication or equivalent confirmation is required.
- Deletion covers roadmap, tasks, progress, chat, and proposals according to retention policy.

- The user receives clear confirmation of irreversible effects.

IAM-006 - Administrative access

Phase: 3 | Priority: P1

Platform administrators MUST use separately authorized access for support and operations.

- Learner data access is least-privilege and auditable.
- AI provider credentials and platform configuration are never exposed to learner roles.

7.2 Learning Profile and Onboarding

LRN-001 - Learning goal capture

Phase: 1 | Priority: P0

The system MUST capture a free-text primary learning goal before roadmap generation.

- The goal is length-limited and content-safety checked.
- Examples may guide specificity without restricting domain.

LRN-002 - Self-assessed level

Phase: 1 | Priority: P0

The learner MUST select a supported starting level without mandatory assessment.

- Initial levels include Complete Beginner, Some Experience, and Intermediate.
- Level values are extensible.

LRN-003 - Learning preferences

Phase: 1 | Priority: P1

The learner MUST select one or more preferred learning modes.

- Initial preferences include hands-on projects, reading/docs, video walkthroughs, and quizzes/drills.
- Preferences influence AI generation but do not forbid other task types.

LRN-004 - Available time

Phase: 1 | Priority: P0

The learner MUST select an available-time preference used for pace estimation.

- Initial options include 15-30 min/day, 30-60 min/day, 1-2 hours/day, and weekends mostly.
- The value is not converted into scheduled calendar events in MVP.

LRN-005 - Desired outcome

Phase: 1 | Priority: P0

The system MUST capture what success looks like for the learner.

- The desired outcome is included in roadmap-generation context.
- The product may show an AI summary for user review before generation.

LRN-006 - Preferred resource language

Phase: 2 | Priority: P0

The learner MUST choose Arabic, English, or Both for external learning resources.

- Resource language is independent from UI language.
- If no valid resource exists in the preferred language, the system may offer a clearly labeled fallback according to policy.

LRN-007 - Onboarding validation

Phase: 1 | Priority: P0

Roadmap generation MUST be blocked until all required onboarding fields are valid.

- Validation errors are field-specific and bilingual.
- Submitted values are persisted before asynchronous generation begins.

7.3 Roadmap Generation and Lifecycle

RMP-001 - Generate roadmap

Phase: 1 | Priority: P0

The system **MUST** generate a personalized roadmap using AI from the persisted learning profile.

- Output is structured, schema-validated, and stored.
- The AI provider response cannot directly write database state.
- Generation stores model/provider/version/correlation metadata for reproducibility and evaluation.

RMP-002 - Roadmap structure

Phase: 1 | Priority: P0

A generated roadmap **MUST** contain ordered stages and ordered tasks with enough metadata to execute the plan.

- Roadmap includes title/summary, estimated duration, status, and version.
- Stages include title, description, order, status, and progress projection.
- Tasks include type, objective, instructions, estimated effort, difficulty, order, and status.

RMP-003 - Generation state machine

Phase: 1 | Priority: P0

Roadmap generation **MUST** use explicit states rather than relying on UI timers.

- Supported states include Draft/InputReady, Generating, Validating, Ready, Failed, Active, Completed, Reset/Archived as applicable.
- Failure stores a safe error category and retry eligibility.

RMP-004 - Schema validation and repair

Phase: 1 | Priority: P0

AI-generated roadmap output **MUST** pass strict server-side validation before becoming Ready.

- Unsupported task types, empty stages, duplicate orders, invalid estimates, or malformed identifiers are rejected.
- The platform **MAY** perform bounded repair/retry before surfacing failure.

RMP-005 - Regenerate before start

Phase: 1 | Priority: P1

The learner **MUST** be able to request a different roadmap before learning activity begins.

- Regeneration creates a new roadmap version or superseding generation rather than mutating silently.
- Once learning begins, major changes use adaptation or reset/rebuild flows.

RMP-006 - Single active roadmap enforcement

Phase: 1 | Priority: P0

The system **MUST** prevent more than one active roadmap per learner in MVP.

- Database/application constraints enforce the rule.
- Schema remains compatible with future multiple roadmaps.

RMP-007 - Goal change rebuild

Phase: 2 | Priority: P0

Changing the primary learning goal after roadmap creation **MUST** require explicit confirmation and rebuild the roadmap.

- Existing progress is not merged into the new plan.
- Old data follows archive/reset retention policy.

RMP-008 - Reset roadmap

Phase: 2 | Priority: P0

The learner **MUST** be able to reset the active roadmap and start over.

- Reset warns about progress impact.
- Reset behavior is distinct from account deletion.

RMP-009 - Roadmap completion

Phase: 2 | Priority: P1

The roadmap **MUST** become Complete when all required stages are Complete.

- Completion timestamp is persisted.
- Completion can unlock a dynamically generated final achievement or summary.

7.4 Stages, Tasks, Completion, and Skip

TSK-001 - Task types

Phase: 1 | Priority: P0

Every task **MUST** use a supported type.

- Initial types: Read, Watch, Quiz, Project, Assignment, Coding Challenge.
- Type-specific metadata may be stored in a flexible validated attributes object or normalized tables.

TSK-002 - Task detail

Phase: 1 | Priority: P0

Learners **MUST** view task objective, instructions, estimated time, difficulty, stage context, and linked resources.

- Task detail identifies whether the task is current, upcoming, complete, skipped, or replaced.
- Instructions render safely without executing untrusted content.

TSK-003 - Mark complete

Phase: 1 | Priority: P0

A learner **MUST** be able to mark an eligible task complete without mandatory proof.

- Completion is idempotent.
- `completed_at` is stored.
- Progress and streak projections update after commit.

TSK-004 - Equal progress weighting

Phase: 1 | Priority: P0

Each required task **MUST** contribute equal weight to progress in MVP.

- Stage progress = completed required tasks / total required tasks.
- Roadmap progress = completed required tasks / total required tasks across active roadmap.
- Skipped/replaced semantics are handled explicitly and not counted as completed by accident.

TSK-005 - Stage completion

Phase: 1 | Priority: P0

A stage **MUST** complete only after all required effective tasks are complete.

- Replaced tasks no longer block stage completion once replacement semantics are committed.
- Future stages become available according to ordering policy.

TSK-006 - Skip request

Phase: 2 | Priority: P0

The learner **MUST** be able to initiate a skip workflow for an accessible task.

- The system captures a reason when configured.
- Mentor receives current task and roadmap context.
- The final user choice may be Keep, Replace, or Skip.

TSK-007 - Replace task

Phase: 2 | Priority: P1

The system **MUST** support AI-proposed replacement of a task when the learner accepts replacement.

- Original task remains traceable.
- Replacement inherits correct stage position and dependencies.
- Replacement resources pass the normal resource pipeline.

TSK-008 - Task ordering

Phase: 1 | Priority: P0

Tasks **MUST** have deterministic ordering within each stage.

- AI may propose reordering later, but user approval is required for material changes.
- Concurrent duplicate order positions are rejected.

7.5 Resource Discovery, Ranking, and Validation

RES-001 - Resource need generation

Phase: 2 | Priority: P0

The roadmap engine SHOULD generate structured resource needs instead of relying on hallucinated URLs.

- Need includes topic, task type, learner level, language, preferred format, and freshness sensitivity.

RES-002 - External discovery

Phase: 2 | Priority: P0

The platform MUST discover candidate resources from supported external search/content sources.

- Discovery is provider-adapter based.
- Queries do not leak unnecessary learner personal data.
- Provider failures degrade gracefully.

RES-003 - Candidate ranking

Phase: 2 | Priority: P0

Candidate resources MUST be ranked using a documented quality policy.

- Factors SHOULD include source authority, relevance, freshness, popularity/engagement where reliable, user sentiment/rating where available, language match, format match, and accessibility.
- No single popularity signal is treated as proof of quality.

RES-004 - URL validation

Phase: 2 | Priority: P0

A recommended resource MUST pass URL and metadata validation before being shown as trusted.

- HTTP availability and redirect destination are checked.
- Unsafe schemes/domains can be blocked.
- last_verified_at and validation status are stored.

RES-005 - Source trust

Phase: 2 | Priority: P0

The platform MUST distinguish authoritative/official sources from community or creator sources.

- Official documentation SHOULD rank highly for fast-changing technical topics where appropriate.
- Community content may be recommended when relevance and quality evidence are strong.

RES-006 - Resource deduplication

Phase: 2 | Priority: P1

The system MUST prevent obvious duplicate resources within a task or stage.

- Canonical URL and normalized title/source signals are used.

RES-007 - Broken/stale resource handling

Phase: 3 | Priority: P1

The platform SHOULD revalidate recommended resources and replace broken or stale resources when detected.

- Broken resources are hidden or clearly marked.
- Replacement is logged without rewriting learner history misleadingly.

RES-008 - Resource provenance

Phase: 2 | Priority: P1

Resource records MUST retain enough provenance to explain why they were recommended.

- Store source/provider, canonical URL, title, type, language, discovered_at, verified_at, and selected ranking signals when available.

7.6 AI Mentor and Context

MNT-001 - Single mentor chat

Phase: 2 | Priority: P0

Each learner MUST have one primary AI Mentor chat stream in MVP.

- Messages persist in chronological order.
- Future per-roadmap or per-task threads remain possible without losing data lineage.

MNT-002 - Context builder

Phase: 2 | Priority: P0

Every mentor request **MUST** use a server-built structured context appropriate to the request origin.

- Base context includes goal, current stage, current task, progress, level, learning preferences, and resource language.
- Screen context may add roadmap, task, resource, or progress details.
- The client cannot forge ownership-sensitive context.

MNT-003 - Context minimization

Phase: 2 | Priority: P0

The platform **MUST** minimize AI context to relevant information required for the response.

- Full raw chat history is not automatically resent indefinitely.
- Summaries or bounded recent-message windows may be used.
- Token budget is measured.

MNT-004 - General Q&A

Phase: 2 | Priority: P0

The mentor **MUST** answer learning questions in context of the active goal and task when relevant.

- It should distinguish explanation from roadmap-changing advice.
- It must not claim a roadmap state that conflicts with persisted data.

MNT-005 - Task assistance

Phase: 2 | Priority: P0

The mentor **SHOULD** help the learner execute the current task without unnecessarily doing the entire learning work for them.

- Guidance may include hints, explanations, examples, debugging, next steps, and resource suggestions.

MNT-006 - Progress-aware recommendations

Phase: 2 | Priority: P1

The mentor **MAY** use persisted progress signals to recommend easier, harder, additional, or replacement work.

- Material roadmap changes are emitted as proposals, not immediate mutations.

MNT-007 - Level change proposal

Phase: 2 | Priority: P1

The mentor **MAY** propose changing the learner level when observed progress strongly suggests mismatch.

- Level changes require explicit learner acceptance.
- Reason is displayed and persisted.

MNT-008 - Tool/action separation

Phase: 2 | Priority: P0

AI natural-language output **MUST** be separated from typed application actions.

- Only allowlisted action schemas can create proposals or resource operations.
- Free-form model text cannot directly invoke arbitrary backend mutations.

MNT-009 - Chat safety and moderation

Phase: 2 | Priority: P0

Mentor input and output **MUST** pass applicable safety and abuse controls.

- Unsafe or disallowed content is handled according to platform policy.
- Safety outcomes are logged without storing unnecessary sensitive detail.

7.7 Roadmap Adaptation Proposals

ADP-001 - Create proposal

Phase: 2 | Priority: P0

The system **MUST** represent material AI-suggested roadmap changes as explicit proposal records.

- Proposal includes type, reason, affected entity, before snapshot, proposed after snapshot, status, model metadata, and timestamps.

ADP-002 - Proposal types

Phase: 2 | Priority: P0

Initial proposal types MUST support add task, replace task, remove/skip task, reorder task, adjust stage content, adjust level, and recommend resource changes.

- New proposal types require schema/policy review.

ADP-003 - Accept proposal

Phase: 2 | Priority: P0

A learner MUST be able to accept a pending proposal.

- Acceptance revalidates that affected roadmap version/state has not changed incompatibly.
- Mutation and proposal status commit transactionally.
- applied_at and resulting roadmap version are recorded.

ADP-004 - Reject proposal

Phase: 2 | Priority: P0

A learner MUST be able to reject a pending proposal without changing the roadmap.

- Rejection preserves the proposal and decision timestamp for traceability.

ADP-005 - Stale proposal handling

Phase: 2 | Priority: P1

The system MUST prevent applying a proposal against incompatible newer roadmap state.

- Stale proposals are marked Expired/Stale and may be regenerated.

ADP-006 - Proposal explanation

Phase: 2 | Priority: P1

Every user-visible proposal SHOULD explain why the change is suggested and what effect it has on the learning path.

- Explanations avoid unsupported certainty about learner capability.

7.8 Progress, Streaks, Milestones, and Achievements

PRG-001 - Overall progress

Phase: 1 | Priority: P0

The system MUST calculate overall roadmap progress from equal-weight effective tasks.

- Calculation is deterministic and server-owned.
- Dashboard and progress screens use the same authoritative formula.

PRG-002 - Stage progress

Phase: 1 | Priority: P0

The system MUST calculate per-stage completed task count and percentage.

- Completed/total counts remain consistent with task state changes.

PRG-003 - Learning timeline

Phase: 2 | Priority: P1

The system SHOULD record major learning events for timeline display.

- Examples: roadmap started, stage started, task completed, stage completed, milestone achieved, proposal accepted, roadmap completed.

PRG-004 - Streak calculation

Phase: 2 | Priority: P0

The system MUST count a day toward streak when at least one task completion occurs in that learner calendar day.

- Timezone handling is explicit.
- Repeated completions on the same day count as one active day.
- Missed days break the streak but do not reschedule the roadmap.

PRG-005 - Dynamic milestones

Phase: 2 | Priority: P1

Milestones **MUST** be generated or derived from the roadmap rather than being limited to a global static list.

- Milestone criteria are persisted in structured form when possible.

PRG-006 - Dynamic achievements

Phase: 2 | Priority: P1

Achievements **MAY** combine roadmap-specific and general progress criteria.

- Unlock events are persisted and idempotent.
- Achievements never alter roadmap correctness.

7.9 Localization, Preferences, and Settings

LOC-001 - Arabic and English UI

Phase: 1 | Priority: P0

The web application **MUST** support Arabic RTL and English LTR as first-class interface languages.

- Layout direction changes globally and safely.
- User-generated text preserves Unicode.

LOC-002 - Independent resource language

Phase: 2 | Priority: P0

Changing UI language **MUST NOT** implicitly overwrite preferred resource language.

- Preferences remain separately persisted.

SET-001 - Edit learning settings

Phase: 2 | Priority: P0

Learners **MUST** be able to edit supported learning preferences and available time.

- Changes that materially affect roadmap structure trigger a warning and adaptation/rebuild policy.
- Non-structural preference changes apply to future mentor/resource behavior.

SET-002 - Edit primary goal

Phase: 2 | Priority: P0

Editing the primary goal **MUST** invoke the confirmed full-roadmap rebuild workflow.

- The system clearly explains that prior roadmap progress will not merge into the new roadmap.

SET-003 - Data controls

Phase: 3 | Priority: P1

Learners **SHOULD** be able to review core account and learning data controls from Settings.

- Account deletion and roadmap reset are clearly distinct destructive actions.

7.10 Platform Administration and AI Governance

ADM-001 - User support lookup

Phase: 3 | Priority: P1

Authorized administrators **MUST** be able to locate users and inspect non-sensitive operational state required for support.

- Access to chat/learning content is separately permissioned and audited.

ADM-002 - AI provider configuration

Phase: 0/3 | Priority: P0

The platform **MUST** centralize AI provider/model configuration behind server-side adapters.

- Credentials are secret-managed.
- Model changes are versioned/configured without client releases where safe.

ADM-003 - Prompt/version governance

Phase: 0/3 | Priority: P0

System prompts, generation contracts, ranking prompts, and action schemas **MUST** be version controlled.

- Each AI artifact stores the effective prompt/config version when practical.

ADM-004 - AI usage telemetry

Phase: 2/3 | Priority: P1

The platform SHOULD measure AI requests, latency, token usage, failures, repair retries, and estimated cost.

- Metrics support provider/model comparison and future pricing decisions.

ADM-005 - Resource governance

Phase: 3 | Priority: P1

Administrators SHOULD be able to block unsafe domains/resources and inspect repeated validation failures.

- Block rules propagate to future recommendations.

ADM-006 - Feature flags

Phase: 3 | Priority: P1

Risky AI capabilities and future features SHOULD be controlled by environment/user cohort flags.

- Flags support safe rollout of new models, proposal types, or resource providers.

8. Non-Functional Requirements

ID	Quality attribute	Acceptance target
NFR-001	Authenticated API latency	p95 <= 500 ms for normal non-AI CRUD/read endpoints under agreed MVP load, excluding third-party latency.
NFR-002	Roadmap generation	User-visible generation SHOULD produce a validated roadmap within 30 seconds p95 under normal provider conditions; longer work exposes clear progress/retry state rather than a frozen UI.
NFR-003	Mentor response	Time-to-first-useful-response SHOULD be <= 8 seconds p95 under normal provider conditions where streaming is supported; full response latency is separately measured.
NFR-004	Resource discovery	Initial verified resource attachment SHOULD complete within 45 seconds p95 or degrade by showing roadmap tasks while resource verification continues asynchronously.
NFR-005	Availability	Production service target 99.9% monthly excluding approved maintenance; third-party AI/search outages are tracked separately.
NFR-006	Durability	No acknowledged task completion, proposal decision, account setting, or roadmap mutation may be lost after successful API acknowledgement.
NFR-007	Recovery	Service RTO target <= 4 hours and data-store RPO target <= 1 hour for severe infrastructure incidents; stronger managed database recovery SHOULD be used where feasible.
NFR-008	Security	TLS 1.2+, modern password hashing, secure secrets, least privilege, rate limiting, CSRF/CORS controls as applicable, and security logging.
NFR-009	Ownership isolation	Every learner-owned roadmap, task, progress, chat, resource association, and proposal query is user-scoped and covered by negative authorization tests.
NFR-010	Localization	Arabic RTL and English LTR are first-class across authentication, onboarding, dashboard, roadmap, tasks, resources,

ID	Quality attribute	Acceptance target
		mentor, progress, and settings.
NFR-011	Accessibility	Core web flows target WCAG 2.1 AA where practicable, including keyboard access, focus visibility, semantic forms, and sufficient contrast.
NFR-012	Compatibility	Support current stable Chrome, Edge, and Firefox on desktop; responsive tablet/mobile web layouts SHOULD remain usable.
NFR-013	Maintainability	Modular monolith boundaries, code review, automated tests, API contracts, migrations, prompt contracts, and ADR/PDR records are mandatory.
NFR-014	Observability	Structured logs, metrics, traces/correlation IDs, queue health, provider latency/error rate, token/cost metrics, generation failures, resource-validation failures, and security events are monitored.
NFR-015	Privacy	Collect the minimum learner data needed; minimize context sent to AI/search providers and document provider retention/data-use settings.
NFR-016	AI traceability	Stored AI-generated roadmaps/proposals SHOULD retain provider/model/config/version metadata sufficient to investigate quality regressions.
NFR-017	AI structured-output reliability	Roadmap/proposal structured outputs MUST pass schema validation; unvalidated model output is never persisted as authoritative state.
NFR-018	Scalability	Stateless APIs, externalized queues/cache, indexed user/roadmap access, and provider adapters allow horizontal scaling without premature microservices.
NFR-019	Cost efficiency	Token/context budgets and caching/summarization strategies SHOULD prevent unbounded per-message AI cost growth as chat history increases.
NFR-020	Safe degradation	If AI or search providers are unavailable, persisted roadmap/progress remains readable and ordinary non-AI state operations continue where safe.
NFR-021	Resource integrity	Recommended external URLs are canonicalized, verified, and protected against unsafe protocols/open-redirect abuse before presentation.
NFR-022	Data portability	Learner data SHOULD be exportable in documented formats in a later control surface without redesigning the data model.

9. Solution Architecture

9.1 Architectural Style

- Modular Monolith backend organized by explicit learning and AI domains with internal application contracts.
- REST API versioned under /api/v1 as the primary frontend integration boundary.
- Synchronous application transactions for authoritative user/roadmap/progress mutations.

- Queues/background workers for roadmap generation, resource discovery/validation, revalidation, analytics, and slow provider work.
- AI-provider and search-provider adapters isolate vendor-specific APIs from domain logic.
- Typed structured-output contracts separate AI reasoning/generation from state mutation.
- CQRS-lite read projections MAY be used for dashboard/progress performance, without a microservice requirement for MVP.
- Persisted database state is the source of truth; AI providers are stateless collaborators from the product perspective.

9.2 Recommended Technology Baseline

Layer	Technology decision	Purpose
Web frontend	React + TypeScript + Vite	Authenticated SPA for onboarding, roadmap, tasks, resources, mentor, progress, and settings; RTL/LTR support.
Backend	Laravel 12 modular monolith on supported PHP 8.3+	Domain services, policies, transactions, queues, APIs, AI orchestration, admin, validation.
Database	MySQL 8.x compatible managed service	Authoritative learner, roadmap, task, chat, progress, proposal, resource, audit, and AI metadata.
Cache / queue	Redis + Horizon/managed workers	Queues, locks, rate limits, short-lived context cache, job visibility.
AI integration	Provider abstraction layer	LLM generation/chat/tool calling with model/version routing and fallback policy.
Resource discovery	Search/content provider adapters	Discover candidate web/video/documentation resources without binding domain logic to one provider.
Object storage	S3-compatible private storage (future-ready)	Exports, optional future assignment uploads, evaluation artifacts; not required for basic MVP text state.
Delivery	Dockerized CI/CD	Repeatable environments, automated tests/migrations, security scanning, rollback, observability.

9.3 Proposed Bounded Contexts

- Identity & Account
- Learning Profile
- Roadmap & Curriculum
- Task Execution
- Progress & Gamification
- Mentor Conversation
- Roadmap Adaptation
- Resource Intelligence
- AI Orchestration & Governance
- Localization & Preferences
- Platform Administration
- Audit & Observability

9.4 Transaction Boundaries

- Mark Task Complete: task state, completion event, progress counters/projection, stage/roadmap transition, streak event, and outbox event commit atomically where correctness depends on them.
- Accept Adaptation Proposal: revalidation, affected roadmap/task mutations, proposal status, roadmap version, and audit history commit atomically.
- Reset/Rebuild Roadmap: active-roadmap transition and replacement generation request are coordinated so the user cannot end with two active roadmaps.

- AI provider calls never hold long database transactions open; request intent is persisted first, provider work occurs outside the transaction, and validated results are committed afterward.
- Resource web checks occur asynchronously and do not block unrelated learner state mutations.

10. Data Architecture

10.1 Identifier Strategy

- Use ULID or UUID identifiers for learner-owned domain entities exposed through APIs.
- Public APIs expose opaque identifiers, never sequential database IDs.
- AI generation requests, proposals, resource validations, and asynchronous jobs use unique correlation/idempotency identifiers.
- Uniqueness constraints enforce one active roadmap per learner in MVP through an application/database strategy compatible with MySQL.

10.2 Major Entities

Domain	Representative entities
Identity	user, session/refresh_token as applicable, password_reset_token, account_deletion_request
Learning Profile	learning_profile, learning_preference, locale_preference, resource_language_preference
Roadmap	roadmap, roadmap_version, stage, task, task_dependency/ordering metadata, roadmap_generation_request
Task Execution	task_completion, task_skip_request, task_replacement_link, task_state_history
Resources	resource, resource_source, task_resource, resource_validation, resource_rank_snapshot
Mentor	mentor_conversation, mentor_message, mentor_context_snapshot, ai_request
Adaptation	adaptation_proposal, adaptation_change, proposal_decision
Progress	progress_snapshot/read model, learning_activity_event, streak_state, milestone, achievement, user_achievement
Platform AI	ai_provider_config, ai_model_config, prompt_version, action_schema_version, feature_flag
Operations	audit_event, outbox_message, job_failure, provider_health_event

10.3 Roadmap State Model

Entity	Representative states
Roadmap	Draft/InputReady -> Generating -> Validating -> Ready -> Active -> Completed; Failed/Reset/Archived as side states
Stage	Upcoming -> Active -> Complete
Task	Upcoming -> Available/Current -> Complete; SkipPending/Skipped/Replaced where applicable
Adaptation Proposal	Pending -> Accepted -> Applied OR Rejected; Stale/Expired when incompatible
Resource	Candidate -> Validating -> Verified -> Active; Broken/Stale/Blocked/Rejected
AI Request	Queued -> Running -> Succeeded; Failed/TimedOut/Cancelled

10.4 Progress Model

- Authoritative task state is used to reconstruct progress; cached percentages are projections for fast reads.
- Effective required task count excludes tasks removed/replaced according to committed change semantics.
- Roadmap progress is not stored as the only truth; it is derivable from task state and may be cached.

- Learning activity events provide the durable basis for streak/timeline calculations.
- Timezone used for streak calculation is stored per user or resolved by explicit platform policy.

10.5 AI and Prompt Data Model

- Persist the minimum provider metadata needed for debugging and quality evaluation: provider, model, model version/alias, prompt/config version, request type, latency, token counts where available, status, and correlation ID.
- Do not store raw hidden reasoning/chain-of-thought. Persist final outputs, structured actions, user-visible explanations, and concise evaluation metadata only.
- Context snapshots MAY store normalized identifiers and summarized state rather than duplicating all roadmap content.
- Prompt text and action schemas live in version-controlled configuration/source and are referenced by version identifiers.

11. AI Orchestration, Context, and Model Strategy

11.1 AI Capability Separation

Capability	Input	Output	State authority
Roadmap Generation	Learning profile + generation policy	Structured roadmap contract	Backend validates and persists
Mentor Chat	User message + compact context	Natural-language answer + optional typed actions	Answer stored; actions require application policy
Adaptation	Roadmap/progress context + user request or AI trigger	Typed change proposal	User accepts/rejects; backend applies
Resource Query Generation	Task/resource need	Search query intents / ranking criteria	Search adapter executes
Resource Ranking	Candidate metadata + learner/task criteria	Ranked candidate decisions	Backend validation decides active recommendation
Progress Interpretation	Progress signals	Advice or proposal	No direct state mutation except through allowed workflows

11.2 Context Construction

- Base context: goal, self-assessed level, learning preferences, available time, desired outcome, resource language, roadmap status, current stage, current task, overall progress.
- Dashboard origin: add next task and recent milestone/streak summary.
- Roadmap origin: add stage/task structure relevant to the question, not necessarily the entire historical plan.
- Task origin: add objective, instructions, task type, task resources, and adjacent task context.
- Resource origin: add selected resource metadata and linked task.
- Progress origin: add stage completion, task counts, streak, milestones, and recent activity events.
- Conversation context uses a bounded recent window plus a compact rolling summary if required; it does not grow without limit.

11.3 Model Routing and Provider Abstraction

- Business logic calls capability interfaces, not vendor SDKs directly.
- Different models MAY be selected for roadmap generation, mentor chat, structured proposals, or summarization based on quality/cost benchmarks.
- Provider/model changes require evaluation against a regression set before broad rollout.
- Fallback models MAY be configured, but fallback must preserve required structured-output/tool capabilities.
- Provider outages produce explicit degraded states; persisted roadmap/progress remains available.

11.4 Structured Output and Tool Safety

- Roadmap generation uses a strict JSON/schema contract with enumerated task types and bounded collection sizes.
- Mentor state-changing intentions are emitted as allowlisted action/proposal objects separate from prose.
- Backend revalidates identifiers, ownership, current versions, and business rules before creating or applying any action.
- AI cannot arbitrarily call database operations, execute URLs, or invent permissions.
- Prompt injection from external resource content is treated as untrusted data and never promoted to system instructions.

11.5 AI Quality Evaluation

- Maintain a representative evaluation set covering programming, design, marketing, languages, business, beginner/intermediate goals, Arabic/English, and different time preferences.
- Roadmap evaluation dimensions: relevance, prerequisite ordering, achievable scope, task clarity, diversity of learning modes, outcome alignment, and estimated-effort realism.
- Mentor evaluation dimensions: contextual correctness, non-contradiction with persisted state, helpfulness, calibrated uncertainty, action-schema validity, and safety.
- Resource evaluation dimensions: relevance, authority, language match, freshness, link validity, and redundancy.
- Production quality metrics and user feedback should be compared by model/prompt version to detect regressions.

12. API and Integration Strategy

12.1 API Principles

- Version REST endpoints under /api/v1.
- Use authenticated ownership-derived user context rather than accepting user_id as authority from the client.
- Use consistent resource responses and machine-readable error envelopes.
- AI jobs return job/request status instead of pretending long provider calls are instant when asynchronous execution is used.
- Idempotency is used for destructive/retry-prone operations such as task completion, proposal acceptance, generation submission, and reset where appropriate.
- OpenAPI documentation is generated and maintained alongside implementation.

12.2 Proposed API Surface

Area	Representative endpoints
Auth	POST /auth/register, /auth/login, /auth/logout, /auth/refresh, /auth/forgot-password, /auth/reset-password, GET /me
Learning Profile	GET/PUT /learning-profile, GET/PUT /preferences
Roadmap	POST /roadmaps/generate, GET /roadmap, POST /roadmap/regenerate, POST /roadmap/reset, PATCH /roadmap/goal
Stages / Tasks	GET /roadmap/stages, GET /tasks/{id}, POST /tasks/{id}/complete, POST /tasks/{id}/skip-request
Resources	GET /tasks/{id}/resources, POST /resources/{id}/report-broken
Mentor	GET /mentor/messages, POST /mentor/messages
Adaptation	GET /adaptation-proposals, POST /adaptation-proposals/{id}/accept, POST /adaptation-proposals/{id}/reject
Progress	GET /progress, GET /progress/timeline, GET /achievements
Settings / Account	PUT /settings, DELETE /account
Admin later	User support, provider health/config, resource blocklist, AI metrics, job failures, feature flags

12.3 External Integrations

Integration	Purpose	Initial policy
LLM provider	Roadmap generation, mentor, proposals, summarization	Server-side only; adapter interface; request minimization; telemetry enabled.
Web/search provider	Discover current external learning resources	Server-side; query based on task need, not personal profile details beyond necessary learning criteria.
Video/content metadata APIs where available	Popularity, recency, duration, channel/source evidence	Optional; absence of popularity data does not invalidate authoritative resources.
Email provider	Registration/password reset transactional email	MVP auth requirement if email recovery is used; learning notifications deferred.

13. Security, Privacy, and AI Safety

13.1 Authentication and Authorization

- Server-side ownership checks protect every learner-owned entity.
- Rate limits apply to login, password reset, AI chat, roadmap generation/regeneration, resource discovery, and expensive endpoints.
- Administrative roles are separate from learner identity and protected with stronger controls; MFA is SHOULD for admin and MAY become MUST before broad public launch.
- Session/token revocation is supported for account deletion and security incidents.

13.2 Privacy

- Only data necessary for personalization is collected in MVP.
- External AI/search requests exclude unnecessary identifiers such as email and internal account IDs.
- Provider privacy, retention, training-use, and regional processing settings are documented before production activation.
- Chat may contain personal information voluntarily supplied by users; retention and deletion semantics apply consistently.
- Account deletion includes provider-related data only to the extent technically and contractually controllable; the platform documents external-provider limitations.

13.3 AI Safety and Trust

- AI output is advisory learning content, not guaranteed professional certification or expert advice.
- The product avoids presenting generated roadmaps as objectively optimal; user control and editable/rebuild flows remain visible.
- External content is untrusted and cannot modify system prompts or tool policy.
- Unsupported URLs, fabricated citations, unsafe downloads, and malicious redirect targets are filtered by resource validation.
- The AI SHOULD state uncertainty when the platform lacks sufficient evidence rather than inventing learner progress or resource facts.
- Safety-sensitive domains may require stricter policy or disclaimer handling before expansion.

13.4 Destructive Actions and Audit

- Roadmap reset, goal-change rebuild, account deletion, proposal acceptance, and administrative overrides create auditable events.
- Application audit records are append-only to ordinary learner roles.
- Logs avoid passwords, reset tokens, full provider secrets, and unnecessary raw sensitive prompt content.

14. Operations, Reliability, and Support

14.1 Environments and Deployment

- Separate development, staging, and production environments with distinct provider credentials and data.
- Containerized deployment with automated database migrations, rollback plan, health checks, and environment validation.
- Feature flags/canary cohorts for new AI models, prompts, resource providers, or proposal capabilities.
- Production model changes are configuration changes with review, not ad-hoc console edits.

14.2 Queues and Background Jobs

- Roadmap generation may execute synchronously or asynchronously depending on provider latency, but job state is durable.
- Resource discovery/validation, revalidation, analytics aggregation, and non-critical AI evaluation run as background jobs.
- Jobs use bounded retries with exponential backoff and dead-letter/failed-job visibility.
- Idempotent job handlers prevent duplicate resource attachment or duplicate roadmap finalization.

14.3 Observability

- Correlation IDs connect API request, AI request, queue job, resource validation, and final persisted change.
- Dashboards track API p95 latency, error rate, AI latency, provider failures, token usage, estimated cost, generation success/repair rate, proposal acceptance, broken-resource rate, and queue backlog.

- Alerts cover database/Redis unavailability, queue failure, provider outage, abnormal AI error rate, authentication abuse, and resource-validation spikes.
- AI quality regressions are investigated by model/prompt version rather than only aggregate product metrics.

14.4 Backup and Recovery

- Automated database backups and point-in-time recovery are configured where supported.
- Restore exercises verify user, roadmap, task, progress, chat, proposal, and resource integrity.
- Configuration/prompt versions are stored in version control and deployable independently of data backups.
- Recovery runbooks define provider-secret rotation and failed-job reconciliation.

14.5 Support

- Initial support includes account recovery, generation failure, incorrect roadmap/resource report, broken link report, and destructive-action recovery policy.
- Learner reports of low-quality AI content should capture roadmap/task/request identifiers and model/prompt metadata without requiring the learner to understand technical details.
- Admin tooling is intentionally built after core learner flows but before broad public scale.

15. Quality Assurance and Release Governance

15.1 Test Strategy

- Unit tests for progress formulas, streak dates, state machines, one-active-roadmap rule, proposal compatibility, and ranking calculations.
- Integration tests for authentication, database transactions, queues, AI provider adapters, search adapters, URL validation, and prompt/action schema parsing.
- Contract tests for /api/v1 and structured AI output schemas.
- End-to-end tests for registration -> onboarding -> generation -> task completion -> stage progress -> mentor -> proposal accept/reject -> reset/delete.
- Negative authorization tests proving one learner cannot access another learner's roadmap, task, chat, progress, or proposal.
- AI regression evaluations across representative learning domains and both languages.
- Resource quality tests using known-good, stale, broken, redirected, duplicate, low-quality, and malicious URL samples.
- Load tests for ordinary API reads plus controlled concurrency for roadmap generation and mentor chat.
- Security tests, dependency scanning, secret scanning, and pre-release penetration testing proportional to launch exposure.
- RTL/LTR visual and functional testing across supported browsers.

15.2 Release Gates

Gate	Minimum evidence
Architecture	Approved ADRs for roadmap lifecycle, one-active-roadmap strategy, AI provider abstraction, structured outputs/actions, resource trust policy, and deletion/retention.
Functional	All P0 acceptance criteria pass; no open critical user-flow defect.
AI Generation	Evaluation set passes agreed relevance/structure thresholds; schema-validation failure and repair rates are within target.
Mentor	Context correctness tests pass; no known route allows model prose to mutate state directly.
Resources	Broken/unsafe URL suite passes; recommended-resource quality review meets agreed threshold.
Security	Ownership negative tests, rate limits, secret handling, dependency scan, and critical vulnerability remediation complete.
Operational	Monitoring, backup/restore, queue visibility, provider outage behavior, runbooks, and rollback rehearsal complete.
Localization	Critical Arabic RTL and English LTR flows pass functional and visual QA.
Pilot	Closed beta feedback collected; critical generation/resource trust defects resolved before wider release.

16. Delivery Plan and Proposed Duration

16.1 Team Assumption

The indicative plan assumes a focused 2-3 developer core team covering backend/AI integration and frontend, with the founder acting as Product Owner and participating in acceptance testing. UX review, DevOps/security consultation, and occasional AI-evaluation/content-quality review improve delivery confidence. A smaller team can deliver the MVP, but provider integration, bilingual QA, and resource verification will compete for the same engineering capacity.

Workstream	Indicative duration	Major deliverables
Foundation and discovery	Weeks 1-3	Final flows, domain model, ADRs, prompt contracts, provider adapters, resource trust policy, evaluation dataset, CI/CD.
Core learner MVP	Weeks 4-8	Auth, onboarding, profile, roadmap generation/persistence, stages/tasks, dashboard, task detail, completion, progress basics, bilingual shell.
Mentor and resources	Weeks 9-13	Context builder, chat, resource discovery/ranking/validation, skip/replace, adaptation proposals, streaks, milestones, achievements.
Hardening and beta	Weeks 14-16	Security, observability, AI evaluations, failure handling, resource revalidation, admin/support minimums, browser/RTL QA.
Controlled MVP release	Weeks 17-18	Closed beta correction, performance tuning, release gates, production rollout, metric baseline.

16.2 Planning Principles

- Roadmap data model and AI structured-output contract are foundation work, not late integration details.
- Resource trust/validation is developed as a product capability, not delegated to prompt wording alone.
- Every AI capability has a deterministic backend boundary and an evaluation method before release.
- Core learner flows ship before sophisticated administration, monetization, notifications, or multiple-roadmap support.
- Every phase ends with a deployable, testable increment and explicit exit criteria.
- Scope changes trade against duration; future features do not silently enter P0 MVP work.

17. Product Metrics

Category	MVP metric
Activation	Percentage of new registered learners who complete onboarding and receive a valid roadmap.
Generation quality	Roadmap generation success rate, schema-repair rate, regenerate-before-start rate, and learner roadmap satisfaction feedback.
Learning engagement	Percentage of activated learners completing at least one task; tasks completed per active learner; stage completion rate.
Retention	D1/D7/D30 active-learning retention and percentage of learners returning after first task completion.
Streak	Median current/longest streak and relationship to task completion retention; treated as behavioral metric, not vanity target.
Mentor value	Mentor usage per active learner, helpfulness feedback, contextual-correction reports, and percent of sessions producing a meaningful next action.
Adaptation	Proposal creation rate, accept/reject rate, stale-proposal rate, and learner completion after accepted adaptation.

Category	MVP metric
Resources	Verified-resource success rate, click/use rate, broken-link report rate, revalidation failure rate, and resource satisfaction.
Reliability	API error rate, AI provider failure rate, queue failure rate, zero acknowledged state-loss incidents.
Performance	p95 API, roadmap generation, mentor response, and resource-validation latency.
AI economics	Tokens and estimated provider cost per activated learner, per roadmap, and per mentor session even while MVP is free.
Localization	Activation/completion metrics segmented by Arabic/English UI and resource-language preference to detect quality gaps.

18. Risks and Mitigations

ID	Risk	Severity	Mitigation
R-01	AI generates poor or incoherent roadmap	Critical	Strict structured schema, evaluation dataset across domains, prompt/model versioning, user regeneration before start, feedback monitoring.
R-02	AI silently corrupts roadmap state	Critical	No direct model writes; typed proposals/actions; backend validation; explicit user approval; transactional mutations.
R-03	Recommended resources are broken, stale, or low quality	Critical	Discovery/validation pipeline, source trust scoring, last_verified_at, revalidation, blacklist, broken-link reporting.
R-04	AI hallucinates URLs or citations	High	Generate resource needs/search intents rather than trusting free-form URLs; verify canonical URL before recommendation.
R-05	Provider outage blocks core product	High	Persisted roadmap remains readable; queues/retries; degraded-mode messaging; provider abstraction/fallback where evaluated.
R-06	AI cost becomes unsustainable while product is free	High	Context minimization, token budgets, usage telemetry, model routing, summarization, caching, rate limits, future entitlement readiness.
R-07	One roadmap per user becomes a schema dead end	High	Model User 1:N Roadmaps but enforce one active roadmap by MVP policy/constraint.
R-08	Progress becomes inconsistent after skip/replacement	High	Explicit task states, replacement lineage, deterministic effective-task formula, invariant tests, transactional proposal application.
R-09	Learner data leaks between accounts	Critical	Ownership-derived queries, policy layer, negative authorization tests, opaque IDs, admin least privilege.
R-10	Prompt injection through external resources	High	Treat web content as untrusted data, separate instructions/data, no arbitrary tool execution,

ID	Risk	Severity	Mitigation
			domain/URL controls.
R-11	Arabic experience is lower quality than English	High	Arabic evaluation cases, RTL QA, localized prompts where needed, resource-language metrics, bilingual acceptance testing.
R-12	Mentor context grows without bound	High	Compact context builder, bounded recent messages, rolling summaries, token budget telemetry.
R-13	Learners over-trust AI recommendations	Medium	User control, calibrated wording, proposal explanations, visible resource source/provenance, avoid claims of guaranteed outcomes.
R-14	Goal change destroys progress unexpectedly	Medium	Explicit confirmation, clear rebuild warning, archive/reset policy, no silent partial merge.
R-15	Admin/support scope delays MVP	Medium	Defer broad admin UI; implement minimum operational controls and observability first.
R-16	Scope expands into assessments, certificates, social, mobile, monetization	High	Explicit non-goals, phase gates, P0/P1/P2 requirements, Product Decision Records for scope changes.

19. Assumptions and Open Validation Items

- AI Mentor is intended for any learning domain, not only programming.
- MVP supports one active roadmap per learner; multiple roadmaps are future scope.
- AI generates the roadmap from scratch in MVP; human-authored templates are future scope.
- Roadmap changes suggested after generation require learner approval before application.
- Learners self-select their starting level; no mandatory placement assessment exists in MVP.
- Ordinary tasks are learner-marked complete without proof; quizzes/projects may exist as task types but are not universal completion gates.
- Resources are external in MVP and must be selected according to quality, relevance, recency/maintenance, trust, and verification policy.
- Mentor context is compact rather than full-history by default.
- One chat stream is used; context changes by entry screen and current roadmap state.
- Available time estimates pace but does not create a calendar schedule; missed days do not reschedule automatically.
- At least one completed task in a learner-local day counts toward streak.
- All tasks have equal progress weight in MVP.
- A stage completes when all effective required tasks are complete.
- Milestones and achievements are dynamic and roadmap-aware.
- Notifications and paid plans are out of MVP scope.
- Admin exists in the product vision but is implemented after core learner capabilities.
- AI-generated roadmap data is persisted; the AI service is not the source of truth.
- Changing the goal rebuilds the roadmap rather than partially merging it.
- Arabic and English are first-class UI languages; preferred resource language is Arabic, English, or Both.
- Target users are self-directed learners including students, employees, professionals, freelancers, career switchers, and people seeking upskilling/reskilling.
- Open validation: exact AI provider/model mix, search/video metadata providers, resource-quality scoring weights, retention period, user timezone source, and beta acceptance thresholds must be confirmed through ADR/PDR or pilot evidence.

Appendix A. Domain Glossary

Term	Definition
Learning Profile	Persisted personalization inputs: goal, level, preferences, time, desired outcome, and language preferences.
Roadmap	The learner's persisted end-to-end learning plan.
Stage	Ordered milestone-level grouping of tasks within a roadmap.
Task	Actionable unit of learning work with type, instructions, estimated effort, and state.
Effective Task	A task that currently counts toward completion after replacements/removals are resolved.
Resource	External learning material attached to a task after discovery and validation.
Mentor Context	Compact structured learner/roadmap state supplied to the AI for a particular request.
Adaptation Proposal	Persisted AI-suggested roadmap change requiring learner accept/reject.
Roadmap Version	Revision identifier used to detect stale proposals and trace major changes.
Generation Request	Durable record of an AI roadmap generation attempt and its status/metadata.
Resource Need	Structured description of what learning material a task requires before a URL is selected.
Streak	Consecutive learner-local days with at least one task completion.
Milestone	Roadmap-specific significant learning checkpoint.
Achievement	Unlocked recognition based on roadmap or general progress criteria.
Prompt Version	Versioned configuration/instruction set used for an AI capability.
Provider Adapter	Application boundary that isolates domain logic from a specific AI/search vendor.
Structured Action	Allowlisted machine-readable AI output that may create a proposal or other controlled application operation.

Appendix B. Requirement Traceability Summary

Product capability	Primary requirements	Primary business rules
Identity & Account	IAM-001..006	BR-025
Onboarding & Profile	LRN-001..007	BR-005, BR-013, BR-016, BR-024
Roadmap Lifecycle	RMP-001..009	BR-001..004, BR-019..021
Tasks & Completion	TSK-001..008	BR-006..012
Resources	RES-001..008	BR-014..016
Mentor	MNT-001..009	BR-004, BR-017, BR-018
Adaptation	ADP-001..006	BR-004
Progress & Gamification	PRG-001..006	BR-008..013, BR-022
Localization & Settings	LOC-001..002, SET-001..003	BR-016, BR-019, BR-024
Platform Governance	ADM-001..006	BR-002, BR-014, BR-023

Appendix C. Confirmed Product Decisions

Decision area	Confirmed MVP decision
Learning domains	Any domain: programming, marketing, design, languages, business, etc.
Roadmaps per learner	One active roadmap in MVP; multiple later.
Generation source	AI generates from scratch in MVP; templates later.

Decision area	Confirmed MVP decision
AI plan changes	AI proposes; learner approves/rejects.
Level	Learner self-selects; no mandatory assessment.
Completion	Mark Complete is sufficient; no proof required.
Task types	Read, Watch, Quiz, Project, Assignment, Coding Challenge.
Resources	External; prioritize trustworthy, relevant, maintained/recent, positively regarded sources and verify links.
Mentor context	Compact Goal + Current Stage + Current Task + Progress + relevant preferences.
Mentor authority	May explain, evaluate, recommend, propose roadmap/tasks/level/resource changes; state changes remain controlled.
Chat	One chat stream; context changes by entry point.
Time	Used to estimate roadmap pace, not calendar scheduling.
Missed days	Roadmap remains unchanged.
Streak	At least one completed task in the day.
Progress weight	Every task has equal weight.
Stage completion	All effective required tasks complete.
Milestones/Achievements	Dynamic according to roadmap.
Notifications	Not in MVP.
Pricing	MVP fully free.
Admin	Required in broader product, implemented after core learner MVP.
AI data persistence	Generated output stored; database is source of truth.
Goal edit	Full roadmap rebuild (Option A).
Languages	Arabic + English; resource language preference Arabic/English/Both.
Skip	Supported; mentor recommends Keep/Replace/Skip.
Regenerate	Allowed before roadmap learning begins.
Destructive controls	Reset Roadmap and Delete Account both supported.
Adaptation record	Store proposal, before/after data, reason, decision, approval/applied timestamps.